Module 04: Cognition — The Robot Brain

Learning Objectives

1. Understand that robots "Think" using **Code** (Algorithms).
2. Learn the **"If-Then"** rule.
3. Know that robots can process information much faster than humans, but they aren't smarter.

Introduction: The Robot Brain

Does a robot dream? Does it have feelings? No. A robot brain is a specialized Computer. It follows a list of instructions called an **Algorithm**. It is like a Recipe:

1. See ball.
2. Grab ball.
3. Throw ball.

If the code says "Eat Ball", the robot will try to eat the ball! It only knows what it is told.

Key Concepts

1. Algorithms (Recipes)

An algorithm is a step-by-step plan.

- **Toothbrush Algorithm**:
    1. Get brush.
    2. Put paste.
    3. Brush teeth.
    4. Spit. If you skip step 2, it doesn't work!

2. If-Then Logic

This is how robots make decisions.

- **IF** it is raining, **THEN** use umbrella.
- **IF** battery is low, **THEN** go to charger. It is a rule for every situation.

3. Loop (Doing it again)

Robots are great at doing boring things. They can tighten a screw 1,000 times perfectly. Humans get tired. Robots (usually) don't!

Activities

Activity 1: The Sandwich Algorithm

Write a recipe for a PB&J Sandwich. Give it to a "Robot" (Friend). If you forgot to say "Open the jar," they can't make it! This shows how precise code must be.

Activity 2: Red Light, Green Light

This game is pure If-Then Logic!

- **IF** Green, **THEN** Run.
- **IF** Red, **THEN** Stop.
- **IF** Yellow, **THEN** Slow down. You are running a traffic algorithm!

Summary

Cognition for a robot means following Code. By using Algorithms and If-Then rules, robots can solve problems and do work without getting confused.

References

- *Hello Ruby: Adventures in Coding* by Linda Liukas
- *How to Code a Sandcastle* by Josh Funk